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Your bill after solar, simulated from the tariff.

Most solar quotes multiply your bill by a saving percentage. The Eternalgy Solar Simulator does the arithmetic instead: it rebuilds your TNB bill from the tariff, follows every kilowatt-hour your system generates, and calculates the bill you will pay after solar, item by item. For homes and for businesses.

Your bill	Tariff match	System size	Energy flow	New bill	Your saving
The ringgit amount and the month on your TNB bill.	The usage that produces that bill under TNB's tariff, with that month's surcharge.	Panels sized to your usage at Malaysian sun-peak hours.	Used on site, exported, or beyond the credit cap.	Every component recalculated at your new usage.	Old bill minus new bill, plus the export credit.

Why the method matters

Solar is a 25-year asset. A saving overstated by RM150 a month is not a small error: it repeats every month the system runs.

Illustration: a RM45,000 system	Saving promised	Saving achieved
Monthly saving	RM600	RM450
Simple payback	6.3 years	8.3 years
20-year saving, no tariff increase	RM144,000	RM108,000
Expectation gap over 20 years		**RM36,000**

For homes: your real bill, rebuilt.

A worked example from the simulator: a RM500 bill from April 2026, three-phase supply, string inverter.

The bill becomes usage

The simulator searches TNB's domestic tariff for the usage that produces RM500 once April's fuel surcharge (−0.47 sen/kWh) is applied: **1,058 kWh a month**, rebuilt as RM499.70.

Energy charge	RM285.98	Network charge	RM135.95
Capacity charge	RM48.14	Retail charge	RM10.00
SST	RM17.08	KWTBB	RM7.52
EEI rebate	RM0.00	Fuel surcharge (AFA)	−RM4.97

The system is sized to that usage

10.4 kWp
16 × 650 W panels

1,061 kWh
a month at 3.4 sun-peak hours

30%
used as it is generated

Every kilowatt-hour is followed

318 kWh is used as it is generated and saves the full tariff. **740 kWh** is exported and credited at RM0.2703/kWh, which is only allowed up to what the home imports that month. **2.8 kWh** is beyond that cap and earns nothing.

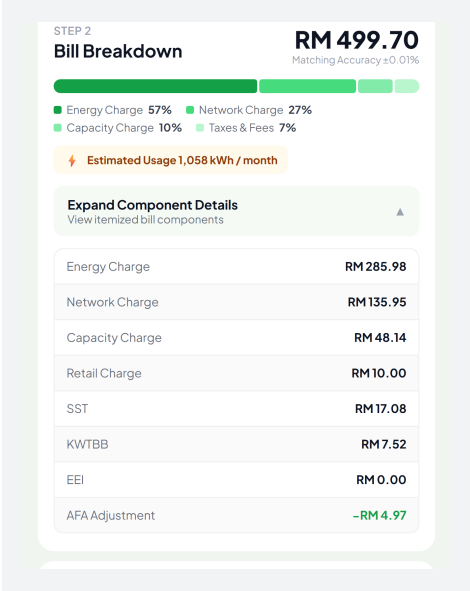
The bill is recalculated, not discounted

Component	Before	After solar
Energy charge	RM285.98	RM199.75
Network charge	RM135.95	RM94.96
Capacity charge	RM48.14	RM33.62
SST, KWTBB and retail	RM34.60	RM14.72
Bill at the new usage	**RM504.67**	**RM343.05**
Export credit, 740 kWh × RM0.2703		−RM199.96
You pay	**RM504.67**	**RM143.09**

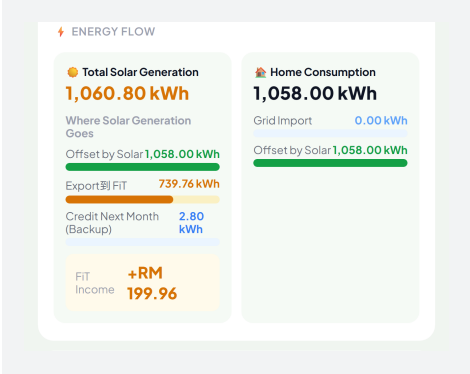
Estimated saving a month

RM361.58

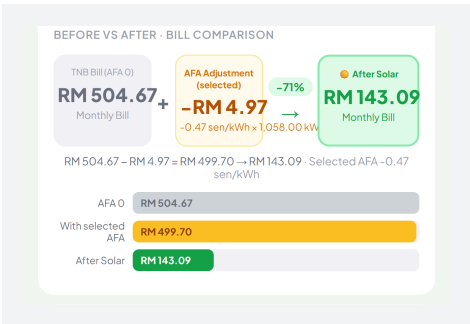
RM504.67 → RM143.09
both at a zero fuel surcharge



The bill matched to 1,058 kWh, component by component.



Where the solar goes, and where the home's power comes from.



Before and after, with the fuel surcharge shown separately.

For business: simulated hour by hour.

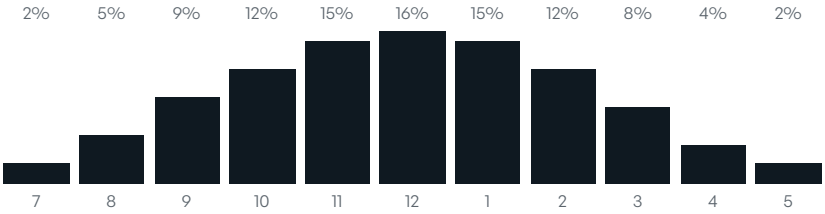
A worked example: a RM5,000 bill on TNB's low-voltage general commercial tariff.

The bill becomes usage

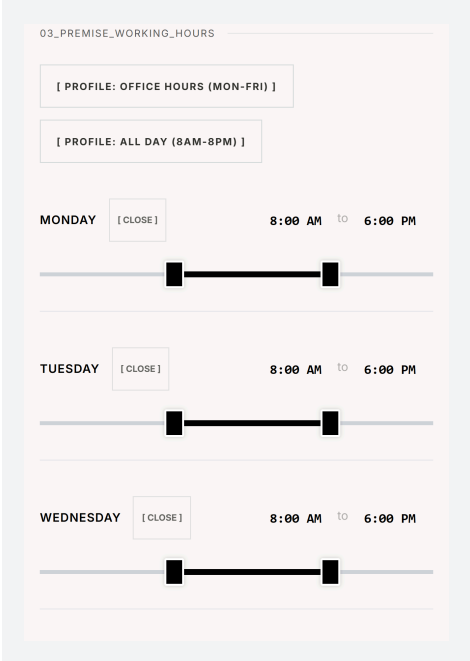
RM5,000 matches **9,671 kWh a month**, rebuilt as RM4,999.68: energy RM2,614.07, capacity RM853.95, network RM1,433.24, retail RM20.00, KWTBB RM78.42.

Your hours shape the result

A business uses most of its power while it is open. The simulator splits the month's usage into a round-the-clock load and an operating load spread over your opening hours, then sets each hour's solar against that hour's load. Solar arrives in a curve that peaks at noon:



Share of a day's solar generated in each hour, 7 am to 5 pm.



Opening hours set for each day of the week.

The system and the week

45.5 kWp	4,601 kWh	88 kWh
70 × 650 W panels	used on site a month	exported a month

The bill is recalculated, not discounted

At the new usage TNB's tariff gives a bill of **RM2,630.07**. The export earns RM17.53 at RM0.20/kWh.

Estimated saving a month

RM2,387.14

RM4,999.68 → RM2,612.54

open every day, 8 am–6 pm

Same bill, closed at weekends

Close the business on Saturday and Sunday and the same panels export most of their weekend output at the lower export rate. The simulator shows the difference instead of hiding it.

Open every day

RM2,387.14

saving a month

Open Monday to Friday

RM1,978.92

saving a month; 1,383 kWh exported



Each day's solar, split into used on site and exported.

Why the numbers hold up.

It starts from the real bill

The bill is matched to the usage that produces it under TNB's current tariff, so the simulation knows the usage instead of guessing it from a percentage.

It applies the bill month's surcharge

The fuel surcharge (AFA) changes every month. Using the one on the bill stops the same ringgit being read as the wrong usage.

It rebuilds the bill after solar

Energy, network, capacity, retail, SST, KWTBB and the efficiency rebate are recalculated at the new usage. Tariff bands and tax thresholds fall away as they would on a real bill.

It follows every kilowatt-hour

Solar used on site saves the full tariff. Exported solar earns only the export rate, only up to what is imported that month; the rest expires, as under ATAP.

It models business hours

Commercial premises are simulated hour by hour against opening hours and round-the-clock load, so a shop closed on Sunday shows a smaller saving.

It shows its working

Every component is on screen, before and after. The customer can check the arithmetic, and so can their bank.

	Percentage estimate	Eternalgy bill simulation
Starting point	Bill × an assumed saving %	**The bill matched to usage under TNB's tariff**
Fuel surcharge	Ignored	**The bill month's AFA applied**
Daytime use and export	Treated as equal	**Valued separately: full tariff vs export rate**
Export limit	Usually ignored	**Credit capped at what is imported; the rest expires**
Bill after solar	Old bill minus the percentage	**Recalculated component by component**
Tax and rebate thresholds	Ignored	**SST, retail charge and EEI follow the new usage**
Business hours	Ignored	**Hour by hour against the opening hours**

What a simulation cannot know

Weather and season, the roof's direction and shading, changes in how much power is used, and future tariffs and rules. The simulator gives a detailed forecast, not a guarantee. We confirm it with a site survey and, ideally, up to twelve months of bills before you commit.

Bring us your TNB bill.
We will show you how the saving is calculated.

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